ABHISHEK KUMAR

AI/ML Engineer | Speech AI & Real-Time Captioning | Computer Vision | OCR | Python
Noida, India | tabhiabhishek@gmail.com | +91 62618 28858 | linkedin.com/in/abhishek-tiwari-5080b4170

PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years of experience building production-grade AI systems across speech recognition, computer vision, and OCR. Currently leading real-time ASR and live broadcast captioning at Planetcast Media Services, delivering on-air captions for live cricket coverage using Whisper, PyTorch, FastAPI, and CUDA-accelerated inference on NVIDIA RTX hardware. Experienced in shipping deep learning models to broadcast/OTT environments, LLM-driven post-processing, and scalable backend microservices.

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Bash, JavaScript
AI/ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, ONNX Runtime, Model Quantization, Transfer Learning
Speech AI / ASR: Whisper (large-v3-turbo), faster-whisper, WhisperX, VAD, CTC Loss, Beam Search, Speaker Diarization
Computer Vision & OCR: OpenCV, YOLOv8, CLIP, SSIM, PaddleOCR (PP-OCRv5), FAISS, Vector Databases
Generative AI / NLP: LLMs, LLM Fine-tuning, Prompt Engineering, RAG, Semantic Validation
Backend & APIs: FastAPI, Flask, Django, REST APIs, WebSockets, Microservices
Streaming & Broadcast: FFmpeg, DeckLink SDK, SDI, UDP/MPEG-TS, HLS, WebRTC, CEA-608/708, VANC
MLOps & Infrastructure: Docker, Git, CI/CD, MLflow, Apache Airflow, NVIDIA CUDA 12.1, RTX A2000, Linux, Windows
Data & Databases: Pandas, NumPy, Matplotlib, Power BI, MongoDB, MySQL

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Planetcast Media Services Limited | Noida, India | *Aug 2025 – Present*

- Leading real-time ASR and live captioning systems for broadcast/OTT using Whisper, PyTorch, and CUDA 12.1, powering on-air captioning for a Cricbuzz client deployment.
- Engineered a cricket-specific semantic validation layer (rule-based + ML) that improved player-name and terminology accuracy by 35%, measured against annotated ground-truth transcripts.
- Built FastAPI microservices with WebSocket streaming delivering CEA-608/708 captions at sub-second latency; integrated SDI capture via DeckLink SDK and real-time VANC caption embedding.
- Reduced end-to-end inference latency by 40% through ONNX Runtime conversion, FP16 quantization, and chunked GPU processing on NVIDIA RTX A2000 hardware.
- Implemented MLOps workflows using MLflow for model versioning, confidence-based fallback routing, and Airflow-orchestrated retraining pipelines.

Freelance AI / Data Analyst | Remote, India | | *Jun 2023 – May 2025*

- Built predictive ML models (churn prediction, demand forecasting) for SMB clients using Scikit-learn, Pandas, and Power BI, enabling data-driven decision-making.
- Automated ETL and data-validation pipelines with Python and SQL, reducing manual data-preparation time by 60%.
- Delivered interactive Power BI dashboards consolidating sales, inventory, and marketing KPIs for executive reporting.

Software Engineer | Advertricssoft IT Systems | Pune, India | *Jan 2021 – May 2023*

- Developed Python automation and monitoring tools for IT incident tracking and root-cause analysis, sustaining 99.5% system uptime with Docker-based deployments.
- Built internal data-extraction scripts and containerized application workflows, forming the foundation for a transition into AI/ML engineering.

Software Engineer | Maxgen Technologies Pvt. Ltd. | Pune, India | *Nov 2019 – Dec 2020*

- Built Python automation frameworks and Django/SQL backend services for large-scale data handling and text mining.
- Optimized SQL queries and structured-extraction tooling, improving data-ingestion throughput by 25%.

KEY AI/ML PROJECTS

CaptionFlow Pro — Real-Time Broadcast Captioning — *PyTorch, Whisper, WhisperX, FastAPI, CUDA, DeckLink SDK, Docker*

- Low-latency ASR pipeline for live cricket with fine-tuned Whisper, WhisperX alignment, hybrid post-processing for player-name correction, 27+ endpoint FastAPI backend with CEA-608 output, and PyInstaller packaging — cutting latency by ~40%.

Live Proxy — Stream Processing & Relaying Engine — *Python, FastAPI, FFmpeg, HLS/WebRTC, Docker*

- Real-time ingest/transcode/relay engine supporting RTMP/SRT/UDP with adaptive bitrate, automatic failover (99.9% uptime), and WebSocket health monitoring — reduced latency by 35%.

DeepQC — AI-Powered Media QC Suite — *OpenCV, YOLOv8, SSIM, FFmpeg, PyInstaller*

- Automated QC pipeline for broadcast files: black-frame, silence, LUFS loudness, A/V sync, SSIM logo detection, and YOLOv8 colorbar classification. Deployed as headless binary.

Additional Projects — *PaddleOCR, CLIP, FAISS, MongoDB, Whisper*

- Sports Scoreboard OCR (real-time broadcast data extraction), AI Visual Search Engine (CLIP + FAISS semantic search), and Automated Sports Highlight Generation (OCR + Whisper + FFmpeg).

EDUCATION

B.E. (Hons.) — Computer Science & Engineering

Lakshmi Narain College of Technology, Bhopal | 2015–2019

CERTIFICATIONS

- Google Data Analytics — Coursera
- SQL for Data Science — Coursera
- Excel for Business — LinkedIn Learning

LANGUAGES

English (Fluent) | Hindi (Native)